

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0559

Orleans 138 kV
September 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0559. The Interconnection Customer proposes a 137.5 MW solar photovoltaic facility interconnecting at the Orleans 138 kV substation in the Pennsylvania area. The request was received on December 31, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 224 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,134,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
COVE AVENUE – ORLEANS 138kV	COVE AVENUE – ORLEANS 138kV #2	115.4	269	52.3
COVE AVENUE – ORLEANS 138kV Line 2	COVE AVENUE – ORLEANS 138kV Line	111.5	279	52.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COVE AVENUE – ORLEANS 138kV	Rebuild line with higher-capacity conductor	\$957,000	100% (\$957,000)
COVE AVENUE – ORLEANS 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,177,000	100% (\$1,177,000)
	Total allocated to project		\$2,134,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.